Pixelcart — Product Document

AI Shopping Agent that replaces browse → filter → search with a single natural language conversation

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1. What We Built

Pixelcart is a conversational AI shopping agent that allows users to describe what they want in plain English, such as:

- "white sneakers under ₹1500"
- "a gift for my mom's birthday"

The system returns ranked, explained product recommendations in a chat-style interface.

Pixelcart is **not a chatbot layered on top of a search bar**. It is a multi-stage AI pipeline consisting of:

Core Pipeline

1. **Intent Extraction**
   Claude AI parses the user's message into structured shopping intent including:
   - Category
   - Budget range
   - Occasion
   - Preferences
   - Recipient
   - Gender
2. **Semantic Search**
   FAISS vector index retrieves products based on meaning rather than keywords.
3. **LLM Re-Ranking**
   Claude re-ranks the retrieved products based on user intent.
4. **Personalised Explanations**
   Each product includes a one-line explanation describing why it fits the query.
5. **Follow-up Question**
   The system asks a smart refinement question to improve results.

Response Time: 2–5 seconds per query

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2. Who It's For

Primary User: Casual Online Shopper

- **Context:** Browsing casually on phone or laptop
- **Behaviour:** Uses natural language instead of filters
- **Pain Point:** Traditional e-commerce requires manual filtering and navigation

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Secondary User: Shopify Store Owners

- **Value:** Adds conversational AI experience to their store
- **Integration Process:**
    - Enter .myshopify.com URL
    - Products auto-sync
    - FAISS index builds
    - Ready to use

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3. Why We Built It

Core Insight

Modern users begin product discovery through AI assistants using natural language. However, when redirected to e-commerce platforms, they are forced back into traditional filter-based interfaces.

PixelCart solves this gap by bringing conversational AI directly into the shopping experience.

Relevance to Kasparro

Kasparro focuses on AI commerce infrastructure. Pixelcart demonstrates:

- How stores can natively support conversational shopping
- What happens when stores "understand" user intent directly

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4. Key Decisions

4.1 Conversational UX over Filters

Decision: No filters, no category navigation, only chat

Reason: Natural language is the core interface

Tradeoff:
Power users searching specific SKUs may find it slower

4.2 Claude Sonnet 4 as Core AI

Decision: Use Claude for:

- Intent extraction
- Re-ranking
- Explanations
- Follow-up generation

Reason: Strong structured output and accuracy

Tradeoff:
4 LLM calls → ~2–4 seconds latency

4.3 FAISS + Sentence Transformers

Decision: Use FAISS with all-MiniLM-L6-v2 embeddings

Reason:

- Captures semantic meaning
- Faster than keyword search
- Works locally

Tradeoff:
Not optimized for very large catalogs (10K+ products)

4.4 Mock Catalogue

Decision: Include 25 curated demo products

Categories Covered:

- Footwear
- Clothing

- Electronics
- Accessories
- Beauty
- Books
- Kids
- Sports

Reason: Enables demo without Shopify

Tradeoff: Static dataset

4.5 Shopify-First Integration

Decision: Support:

- Storefront GraphQL API
- Public REST API

Reason: Maximizes compatibility across stores

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5. Scope

Included (MVP)

- Natural language product search
- Multi-turn conversation
- AI-based intent extraction
- FAISS semantic search
- LLM re-ranking
- Personalised explanations
- Follow-up questions
- Shopify integration (optional)
- Glassmorphism UI (dark mode)
- Demo mode

Excluded (Intentional)

Feature	Reason
Add-to-cart / checkout	Focus is discovery
User authentication	Demo-first
Order tracking	Not relevant
Multi-store comparison	Adds complexity
Payment processing	Shopify handles it
Product reviews	Requires extra data
Mobile app	Web-first approach

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6. Walkthrough

Step 1: Setup Screen

- Left panel:
  - Brand identity
  - “How it works”
  - Example queries
  - Tech stack
- Right panel:
  - Shopify URL input
  - Backend health check
  - Launch/Demo buttons

Step 2: Connect or Demo

- **With Shopify:**
  - Enter store URL
  - Products fetched
  - FAISS index built

- **Demo Mode:**
    - Load mock catalog
    - Start immediately

Step 3: Conversational Shopping

- User enters query
- Intent pills appear
- Product cards displayed with:
    - Image
    - Price
    - Rating
    - Explanation
    - Tags
- Follow-up question is generated
- Response time displayed

Step 4: Refinement

- User adds more details
- Context retained
- Results updated dynamically

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7. Success Metrics

Metric	Target	Measurement
Query Satisfaction	80%+	User clicks/feedback
Response Latency	< 5 sec	Backend logs
Intent Accuracy	90%+	Manual evaluation
Session Depth	2.5+ queries	Analytics

Metric	Target	Measurement
Shopify Success Rate	95%+	Error tracking

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8. Future Directions

1. Streaming results (real-time updates)
2. Voice-based input
3. Multi-store aggregation (Amazon, Flipkart, Shopify)
4. Personalisation memory
5. Chat-based checkout
6. Merchant analytics dashboard